

Hamidreza Eivazi

Postdoctoral Researcher in Scientific Machine Learning

📍 Braunschweig, Germany | ✉ h.eivazi@tu-braunschweig.de
🌐 hamidrezaeiv.github.io | 📄 github.com/HamidrezaEiv | 📞 0000-0003-3650-4107 | 🎓 Google Scholar

PROFILE

Postdoctoral researcher at the Institute of Applied Mechanics, Technische Universität Braunschweig, working at the intersection of scientific and physics-informed machine learning, operator learning, multiscale mechanics, turbulence, and battery degradation modeling. Develops research software for AI-accelerated simulation, with experience spanning neural operators, reduced-order modeling, generative AI, diffusion models, and HPC-enabled workflows.

APPOINTMENTS

Institute of Applied Mechanics, Technische Universität Braunschweig

2026 – Present

Postdoctoral Researcher, Division Data-Driven Modeling of Mechanical Systems

Braunschweig, Germany

- Physics-informed machine learning for computational mechanics.

ISSE, TU Clausthal; CircularLiB, TU Braunschweig

2021 – 2025

Research Assistant and PhD Student; CircularLiB member

Clausthal-Zellerfeld /
Braunschweig, Germany

- Developed EquiNO, a physics-informed neural operator for multiscale simulations.
- Built surrogate constitutive models for FE^2 workflows and integrated them into GPU-, MPI-, and JIT-enabled software pipelines.
- Developed explainable and generative AI methods for Li-ion battery cycle-life prediction, including DiffBatt.
- Advanced operator learning and nonlinear model reduction, leading to publications in JCP, ICLR, NeurIPS workshop, and PAMM.

KTH Royal Institute of Technology

2019 – 2021

Visiting Researcher

Stockholm, Sweden

- Developed PINNs for RANS equations and physics-informed enhancement of PIV measurements.
- Worked on Koopman and recurrent neural network models, as well as autoencoder-based reduced-order models for turbulence.
- Contributed to applied machine-learning projects in climate-action NLP and thunderstorm prediction.

University of Tehran

2017 – 2019

Researcher

Tehran, Iran

- Developed deep-learning methods for nonlinear model reduction and unsteady flow simulation.
- Introduced hybrid dynamic mode decomposition and machine-learning-based turbulence forecasting.

EDUCATION

Technische Universität Clausthal

PhD in Scientific Machine Learning and Multiscale Simulation; Summa cum laude

Clausthal-Zellerfeld, Germany

University of Tehran

M.Sc. in Aerospace Engineering, major in Aerodynamics

Tehran, Iran

University of Mazandaran

B.Sc. in Mechanical Engineering, major in Mechanics

Mazandaran, Iran

SELECTED PUBLICATIONS

Full list: Google Scholar

Hamidreza Eivazi, J.-A. Tröger, S. Wittek, S. Hartmann, & A. Rausch (2026). EquiNO: A Physics-Informed Neural Operator for Multiscale Simulations. *Journal of Computational Physics*, 554, 114745.

S. Dhankhar, S. Wittek, Hamidreza Eivazi, & A. Rausch (2025). R2RNet: A deep spatiotemporal Rain to River Network for water level prediction and flood forecasting. *Journal of Hydrology: Regional Studies*, 61, 102571.

Hamidreza Eivazi, A. Hebenbrock, R. Ginster, S. Blömeke, S. Wittek, S. Hermann, T. S. Spengler, T. Turek, & A. Rausch (2024). DiffBatt: A Diffusion Model for Battery Degradation Prediction and Synthesis. In *NeurIPS 2024 Workshop on Foundation Models for Science*.

Hamidreza Eivazi, M. Alikhani, J.-A. Tröger, S. Wittek, S. Hartmann, & A. Rausch (2024). Enhancing Multiscale Simulations with Constitutive Relations-Aware Deep Operator Networks. *PAMM*, e202400123.

Hamidreza Eivazi, S. Wittek, & A. Rausch (2024). Nonlinear model reduction for operator learning (KPCA-DeepONet). In *The Second Tiny Papers Track at ICLR 2024*.

Hamidreza Eivazi, Y. Wang, & R. Vinuesa (2024). Physics-informed deep-learning applications to experimental fluid mechanics. *Measurement Science and Technology*, 35(7), 075303.

Hamidreza Eivazi, J.-A. Tröger, S. Wittek, S. Hartmann, & A. Rausch (2023). FE² Computations with deep neural networks: Algorithmic structure, data generation, and implementation. *Mathematical and Computational Applications*, 28(4).

G. Hasanuzzaman, Hamidreza Eivazi, S. Merbold, C. Egbers, & R. Vinuesa (2023). Enhancement of PIV measurements via physics-informed neural networks. *Measurement Science and Technology*, 34(4), 044002.

Hamidreza Eivazi, M. Tahani, P. Schlatter, & R. Vinuesa (2022). Physics-informed neural networks for solving Reynolds-averaged Navier–Stokes equations. *Physics of Fluids*, 34(7).

Hamidreza Eivazi, S. Le Clainche, S. Hoyas, & R. Vinuesa (2022). Towards extraction of orthogonal and parsimonious non-linear modes from turbulent flows. *Expert Systems with Applications*, 117038.

Hamidreza Eivazi, L. Guastoni, P. Schlatter, H. Azizpour, & R. Vinuesa (2021). Recurrent neural networks and Koopman-based frameworks for temporal predictions in a low-order model of turbulence. *International Journal of Heat and Fluid Flow*, 90, 108816.

Hamidreza Eivazi, H. Veisi, M. H. Naderi, & V. Esfahanian (2020). Deep neural networks for nonlinear model order reduction of unsteady flows. *Physics of Fluids*, 32(10), 105104.

SELECTED SOFTWARE

EquiNO – github.com/HamidrezaEiv/EquiNO

– Physics-informed neural operators for multiscale simulations.

DiffBatt – github.com/HamidrezaEiv/DiffBatt

– Diffusion models for Li-ion battery degradation prediction and synthetic data generation.

KPCA-DeepONet – github.com/HamidrezaEiv/KPCA-DeepONet

– Kernel PCA and DeepONet framework for operator learning.

PINNs for RANS – github.com/KTH-FlowAI/PINNs-for-RANS

– PINN-based solver for Reynolds-averaged Navier–Stokes equations.

TEACHING AND SERVICE

- Lecturer for *Applied Deep Learning* and *Introduction to Artificial Intelligence*, Technische Universität Clausthal (2023–2024).
- Teaching Assistant for *Data-driven Methods in Engineering*, KTH Royal Institute of Technology/University of Bologna (2020–2021).
- Supervised thesis projects on battery degradation prediction, neural operators for FE² computations, and turbulence forecasting.
- Reviewer for *Journal of Computational Physics*, *Physics of Fluids*, *International Journal of Heat and Fluid Flow*, *Physical Review Fluids*, *Scientific Reports*, *Data in Brief*, *Soft Computing* and related venues.

SELECTED TALKS AND CONFERENCES

- 2026 – 14th GAMM AG Data Workshop, TU Braunschweig; 96th GAMM Annual Meeting, Stuttgart, Germany.
- 2025 – Workshop on Simulation Sciences 2025, Clausthal-Zellerfeld, Germany; 95th GAMM Annual Meeting, Poznan, Poland.
- 2024 – NeurIPS 2024, Vancouver, Canada; ICLR 2024, Vienna, Austria; 94th GAMM Annual Meeting, Magdeburg, Germany.

RESEARCH INTERESTS

AI for Science, Scientific Machine Learning, Operator Learning, Physics-Informed ML, Multiscale Mechanics, Turbulence Modeling, Battery Degradation Modeling, Research Software, High-Performance Computing

TECHNICAL SKILLS

Python, JAX, PyTorch, TensorFlow, Matlab, Julia, Fortran, MPI, CUDA, OpenMP, FEniCS, OpenFOAM, Docker, Git, LaTeX